Energy use fell. Was it efficiency?

The estimand

The change in EU27 final energy consumption of industry, services and agriculture between 2015 and 2024, in Mtoe, split by additive LMDI-I into an activity effect, a structure effect and an intensity effect that sum to the observed change exactly.

The answer

Consumption fell 26.1 Mtoe (-6.7% of 387 Mtoe), 95% interval [-57.7, -3.7].
Activity added 60.4 Mtoe [39.9, 84.9].
Structure removed 16.5 Mtoe [-27.7, -7.5]: value added moved to leaner cells.
Intensity removed 70.0 Mtoe [-112.3, -37.1], -18.1% of 2015 use.

Intensity is 81% of the offset to activity growth [70%, 89%].

What it does not say

73% of the intensity effect falls in 2022 and 2023, when the real energy price index (HICP energy over all items, 2015=100) went from 104 to 129. A fall in energy per euro of value added is not th as an efficiency gain, and the deck shows where the two come apart.

Nothing here is a forecast

The decomposition accounts for a change that has happened. It projects nothing, so there is no backtest; the checks are the identity, the sweep and Eurostat's own series.

Method

Data

Eurostat nrg_bal_s (final consumption by sector, KTOE, all fuels) and nama_10_a10 (gross value added, chain-linked 2015 prices), pulled live from the public API by src/fetch.py. 27 member states, three sectors each: 81 cells. Households and transport are outside the estimand because they have no value-added denominator.

Sector mapping

Industry (FC_IND_E) over NACE B-E and F; services (FC_OTH_CP_E) over G to U; agriculture (FC_OTH_AF_E) over A. The mapping is a choice; the sweep varies it.

Decomposition

For each cell i: E = Q x S_i x I_i, with Q total value added, S_i the cell's share and I_i its energy per unit of value added. Additive LMDI-I weights each log ratio by the logarithmic mean of the cell's energy in the two years, which leaves no residual. Effects are computed year by year and summed (chained).

Interval

Cluster bootstrap over member states, 2000 draws, seed 14. A state comes with all three of its sectors; the total, activity and structure are recomputed on every draw.
It answers: how much does the headline depend on which states the EU is made of.
It does not measure error in Eurostat's figures, which the identity cannot see.

Check on the arithmetic

Identity residual 7e-11 ktoe; largest over 2000 bootstrap draws 6e-10 ktoe.

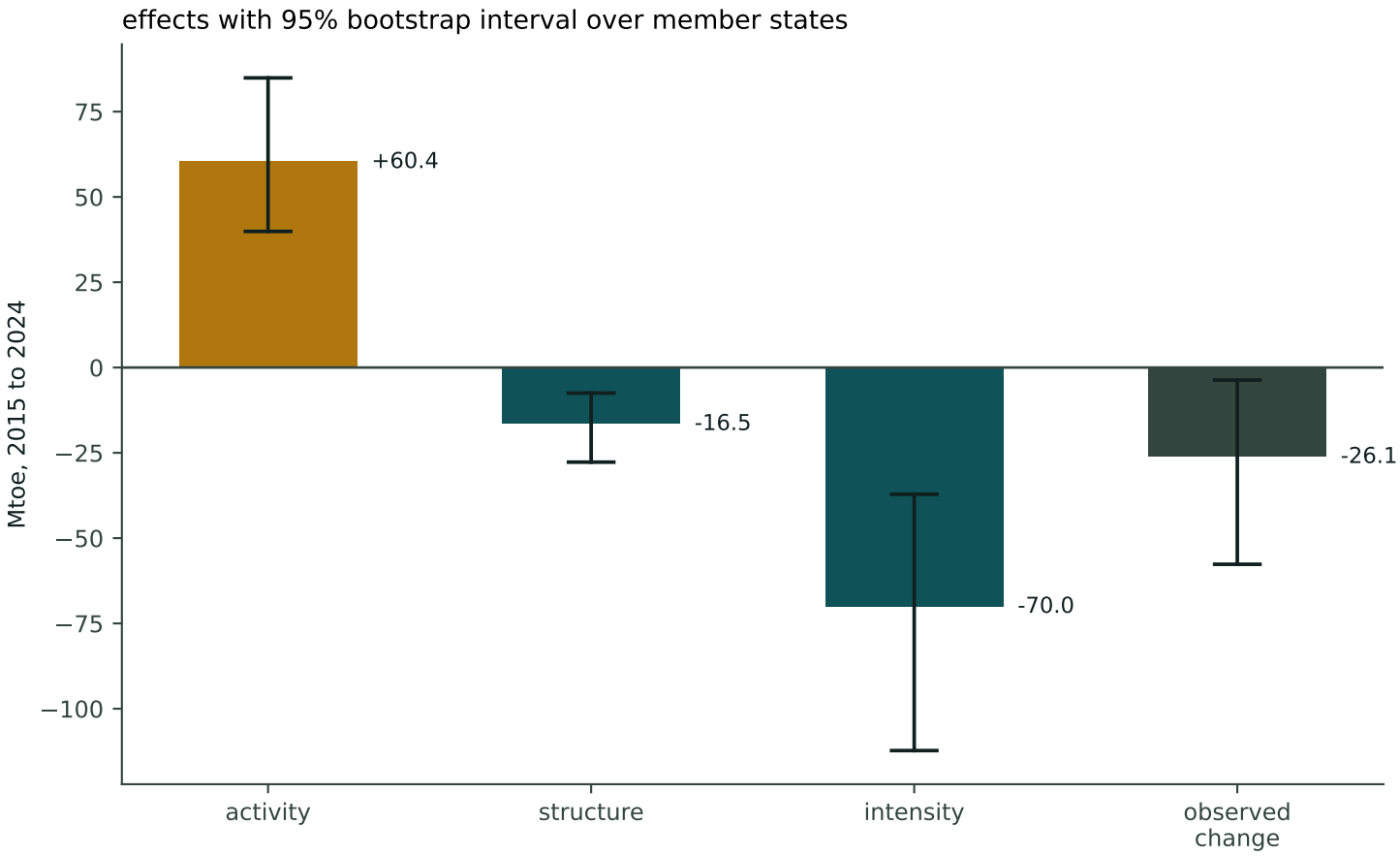
Result

Three effects, one identity

Observed -26.1 Mtoe = activity 60.4 + structure -16.5 + intensity -70.0.
As a share of 2015 use: +15.6%, -4.3%, -18.1%, net -6.7%.

What the intervals support

Intensity is negative in 100% of draws and structure in 100%.
The net fall is below zero across its interval [-10.7%, -1.3%].
Intensity exceeds activity growth in 76% of draws: the point estimates say yes, the interval does not settle it. Do not read the two bars as a ranking.
Germany is the most influential state: without it the intensity effect is -51.7 Mtoe, not -70.0.



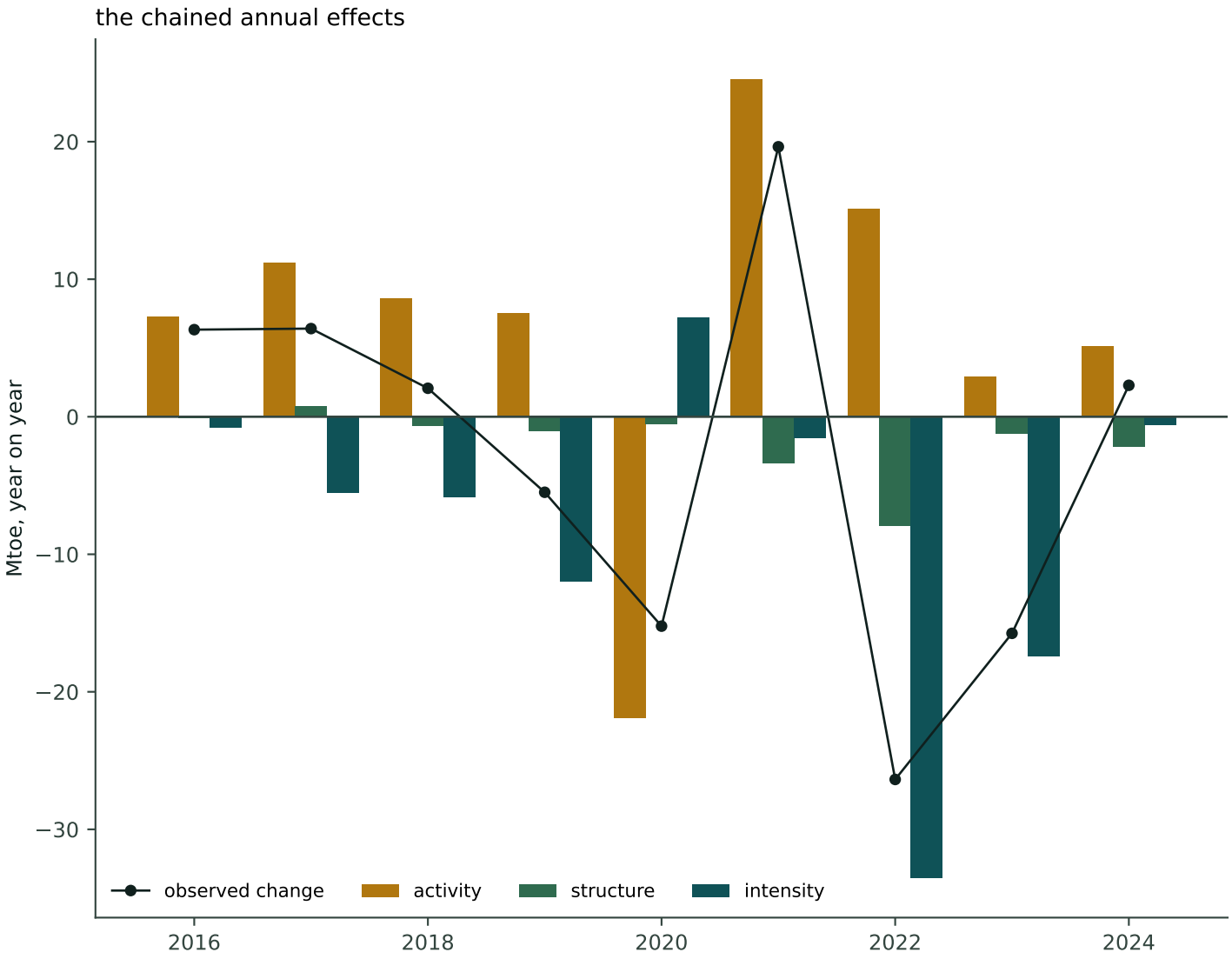
When it happened, and where

The timing is the finding

2016 to 2019: intensity effects sum to -24.2 Mtoe; activity added +34.6.
2020: activity -21.9, intensity +7.2. Output fell faster than energy, so intensity rose.
2022 and 2023: intensity -33.5 and -17.4 Mtoe; real energy price index 104, 129, 121 (2021 to 2023).
Those two years carry 73% of the 2015 to 2024 intensity effect [62%, 88%].

By sector (Mtoe)

Industry: intensity -40.2, structure -15.4, activity 36.5.
Services: intensity -31.9, structure 3.6, activity 19.6.
Agriculture: intensity 2.1, structure -4.6, activity 4.2.
Sectors whose intensity rose: agriculture.



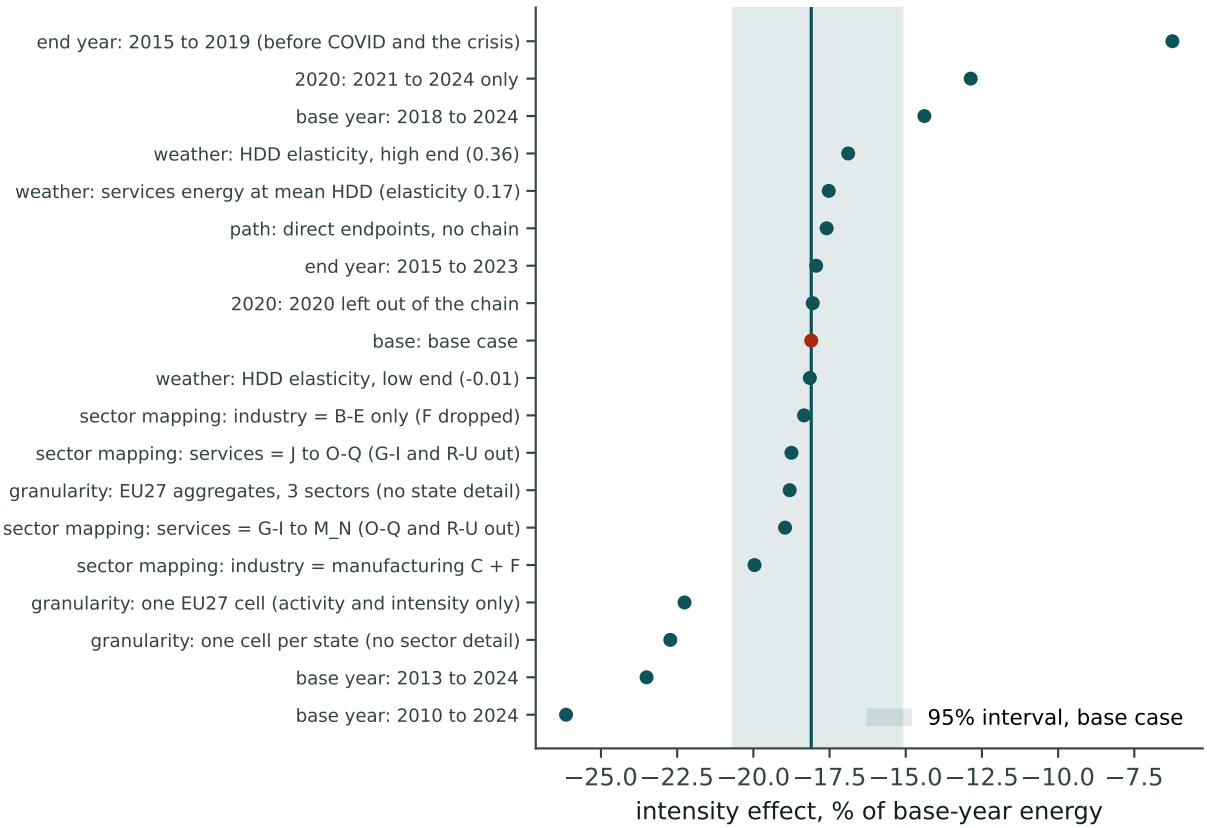
Validation and sensitivity

Against what Eurostat publishes

Energy summed over states equals the published EU27 total in every sector and year; value added differs by at most 0.75% (chain-linked volumes are not strictly additive). Final energy per unit of value added of the productive economy: -20.1%. Eurostat's own energy intensity of the economy (nrg_ind_ei, gross available energy per GDP): -22.2%. Different numerators and denominators, same direction and order of magnitude; no like-for-like series exists.

What moves the intensity effect

Across 19 variants it runs from -26.1% to -6.3% of base-year energy; it is negative in all.
Sector mapping: -20.0% to -18.3%. Base year: -26.1% to -14.4%.
End year: -6.3% to 2019, -12.9% for 2021-2024 alone. 2020 in or out of the chain: no change.
Weather (services, HDD elasticity 0.17, se 0.10): -18.1% to -16.9%.
Granularity matters most for structure: +0.4% with one cell per state, -3.5% with EU27 aggregates only, -4.3% with both dimensions.



Limits, and what I would build next

The assumption that breaks it

That a fall in energy per unit of value added is efficiency. It is an accounting residual: fuel switching, price-driven curtailment and lower output of energy-intensive products inside a sector all land in it. 73% of it sits in 2022 and 2023, which is consistent with curtailment, though nothing here proves it. Three sectors cannot separate the cases.

Sector detail

Moving from one cell per state to three sectors shifted 4.6 points from intensity to structure. Finer industry detail may shift more; I have not tested it, so the intensity effect here is not a clean estimate of within-activity change.

Other attacks

Weather: EU27 heating degree days fell 7.0% between 2015 and 2024; correcting services energy moves the intensity effect by at most 1.2 points, and the elasticity is itself estimated. Value added is a volume index: structural change in prices and the treatment of energy-sector own use are not modelled. The interval covers the choice of states, not revisions to Eurostat's figures.

Next

Industry by branch (nrg_bal_c against nama_10_a64) to split structure from intensity where it matters, and the same decomposition on 2022 to 2024 with monthly prices.